Volume 03 - Book 03.04 Knowledge System

Version v6 draft

README.md

Book 03.04 - Knowledge System

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-README

Purpose

Define the Knowledge System blueprint package and its subsystem decomposition as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for the Knowledge System blueprint package and its subsystem decomposition. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for the Knowledge System blueprint package and its subsystem decomposition.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for the Knowledge System blueprint package and its subsystem decomposition.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.

- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeSystemDefined
- KnowledgeSystemIngested
- KnowledgeSystemIndexed
- KnowledgeSystemReviewed
- KnowledgeSystemSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge System has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

INDEX.md

Book 03.04 Index

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-INDEX

Purpose

Define the canonical reading order and document IDs for Knowledge System as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for the canonical reading order and document IDs for Knowledge System. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for the canonical reading order and document IDs for Knowledge System.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for the canonical reading order and document IDs for Knowledge System.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- Book0304IndexDefined
- Book0304IndexIngested
- Book0304IndexIndexed
- Book0304IndexReviewed
- Book0304IndexSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Book 03.04 Index has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011

- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.01 - Knowledge System Overview

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-001

Purpose

Define the governed knowledge layer that captures, validates, links, searches, versions, secures, and serves enterprise knowledge as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for the governed knowledge layer that captures, validates, links, searches, versions, secures, and serves enterprise knowledge. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for the governed knowledge layer that captures, validates, links, searches, versions, secures, and serves enterprise knowledge.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for the governed knowledge layer that captures, validates, links, searches, versions, secures, and serves enterprise knowledge.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeSystemOverviewDefined
- KnowledgeSystemOverviewIngested
- KnowledgeSystemOverviewIndexed
- KnowledgeSystemOverviewReviewed
- KnowledgeSystemOverviewSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge System Overview has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.02 - Knowledge Graph

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-002

Purpose

Define the graph of entities, relationships, provenance, claims, topics, owners, and domain concepts as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for the graph of entities, relationships, provenance, claims, topics, owners, and domain concepts. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for the graph of entities, relationships, provenance, claims, topics, owners, and domain concepts.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for the graph of entities, relationships, provenance, claims, topics, owners, and domain concepts.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeGraphDefined
- KnowledgeGraphIngested
- KnowledgeGraphIndexed
- KnowledgeGraphReviewed
- KnowledgeGraphSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Graph has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.03 - Knowledge Store

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-003

Purpose

Define the canonical storage boundary for approved, draft, archived, and superseded knowledge objects as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for the canonical storage boundary for approved, draft, archived, and superseded knowledge objects. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for the canonical storage boundary for approved, draft, archived, and superseded knowledge objects.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for the canonical storage boundary for approved, draft, archived, and superseded knowledge objects.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeStoreDefined
- KnowledgeStoreIngested
- KnowledgeStoreIndexed
- KnowledgeStoreReviewed
- KnowledgeStoreSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Store has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.04 - Document Intelligence

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-004

Purpose

Define document understanding, classification, segmentation, metadata extraction, and evidence preservation as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for document understanding, classification, segmentation, metadata extraction, and evidence preservation. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for document understanding, classification, segmentation, metadata extraction, and evidence preservation.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for document understanding, classification, segmentation, metadata extraction, and evidence preservation.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- DocumentIntelligenceDefined
- DocumentIntelligenceIngested
- DocumentIntelligenceIndexed
- DocumentIntelligenceReviewed
- DocumentIntelligenceSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Document Intelligence has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.05 - Memory System

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-005

Purpose

Define working, episodic, semantic, user, team, organization, and agent memory boundaries as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for working, episodic, semantic, user, team, organization, and agent memory boundaries. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for working, episodic, semantic, user, team, organization, and agent memory boundaries.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for working, episodic, semantic, user, team, organization, and agent memory boundaries.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- MemorySystemDefined
- MemorySystemIngested
- MemorySystemIndexed
- MemorySystemReviewed
- MemorySystemSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Memory System has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011

- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.06 - Vector Store

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-006

Purpose

Define vector persistence, embedding references, similarity search constraints, lifecycle, and isolation as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for vector persistence, embedding references, similarity search constraints, lifecycle, and isolation. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for vector persistence, embedding references, similarity search constraints, lifecycle, and isolation.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for vector persistence, embedding references, similarity search constraints, lifecycle, and isolation.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- VectorStoreDefined
- VectorStoreIngested
- VectorStoreIndexed
- VectorStoreReviewed
- VectorStoreSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Vector Store has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.07 - Embedding System

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-007

Purpose

Define embedding generation, model selection, versioning, refresh, quality checks, and provider independence as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for embedding generation, model selection, versioning, refresh, quality checks, and provider independence. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for embedding generation, model selection, versioning, refresh, quality checks, and provider independence.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for embedding generation, model selection, versioning, refresh, quality checks, and provider independence.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- EmbeddingSystemDefined
- EmbeddingSystemIngested
- EmbeddingSystemIndexed
- EmbeddingSystemReviewed
- EmbeddingSystemSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Embedding System has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.08 - Semantic Search

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-008

Purpose

Define semantic retrieval, hybrid search, ranking, filtering, explanations, and query governance as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for semantic retrieval, hybrid search, ranking, filtering, explanations, and query governance. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for semantic retrieval, hybrid search, ranking, filtering, explanations, and query governance.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for semantic retrieval, hybrid search, ranking, filtering, explanations, and query governance.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- SemanticSearchDefined
- SemanticSearchIngested
- SemanticSearchIndexed
- SemanticSearchReviewed
- SemanticSearchSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Semantic Search has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011

- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.09 - Knowledge Indexing

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-009

Purpose

Define index construction, incremental updates, invalidation, freshness, and query-time routing as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for index construction, incremental updates, invalidation, freshness, and query-time routing. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for index construction, incremental updates, invalidation, freshness, and query-time routing.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for index construction, incremental updates, invalidation, freshness, and query-time routing.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeIndexingDefined
- KnowledgeIndexingIngested
- KnowledgeIndexingIndexed
- KnowledgeIndexingReviewed
- KnowledgeIndexingSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Indexing has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011

- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.10 - Knowledge Versioning

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-010

Purpose

Define version history, supersession, rollback, diffing, release references, and compatibility as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for version history, supersession, rollback, diffing, release references, and compatibility. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for version history, supersession, rollback, diffing, release references, and compatibility.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for version history, supersession, rollback, diffing, release references, and compatibility.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeVersioningDefined
- KnowledgeVersioningIngested
- KnowledgeVersioningIndexed
- KnowledgeVersioningReviewed
- KnowledgeVersioningSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Versioning has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011

- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.11 - Knowledge Ingestion

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-011

Purpose

Define source onboarding, connectors, upload flows, validation, staging, and ingestion governance as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for source onboarding, connectors, upload flows, validation, staging, and ingestion governance. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for source onboarding, connectors, upload flows, validation, staging, and ingestion governance.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for source onboarding, connectors, upload flows, validation, staging, and ingestion governance.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeIngestionDefined
- KnowledgeIngestionIngested
- KnowledgeIngestionIndexed
- KnowledgeIngestionReviewed
- KnowledgeIngestionSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Ingestion has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.12 - Knowledge Extraction

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-012

Purpose

Define structured extraction of claims, entities, facts, tasks, decisions, obligations, and references as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for structured extraction of claims, entities, facts, tasks, decisions, obligations, and references. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for structured extraction of claims, entities, facts, tasks, decisions, obligations, and references.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for structured extraction of claims, entities, facts, tasks, decisions, obligations, and references.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeExtractionDefined
- KnowledgeExtractionIngested
- KnowledgeExtractionIndexed
- KnowledgeExtractionReviewed
- KnowledgeExtractionSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Extraction has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010
- MAL-V00-B011

- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.13 - Knowledge Normalization

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-013

Purpose

Define canonical forms, deduplication, taxonomy alignment, confidence handling, and schema conformity as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for canonical forms, deduplication, taxonomy alignment, confidence handling, and schema conformity. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for canonical forms, deduplication, taxonomy alignment, confidence handling, and schema conformity.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for canonical forms, deduplication, taxonomy alignment, confidence handling, and schema conformity.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeNormalizationDefined
- KnowledgeNormalizationIngested
- KnowledgeNormalizationIndexed
- KnowledgeNormalizationReviewed
- KnowledgeNormalizationSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Normalization has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.14 - Knowledge Linking

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-014

Purpose

Define relationship creation, citation links, dependency links, ownership links, and cross-volume references as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for relationship creation, citation links, dependency links, ownership links, and cross- volume references. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for relationship creation, citation links, dependency links, ownership links, and cross-volume references.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for relationship creation, citation links, dependency links, ownership links, and cross-volume references.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeLinkingDefined
- KnowledgeLinkingIngested
- KnowledgeLinkingIndexed
- KnowledgeLinkingReviewed
- KnowledgeLinkingSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Linking has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.15 - Knowledge Quality

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-015

Purpose

Define quality scoring, freshness, trust, review states, evidence requirements, and correction loops as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for quality scoring, freshness, trust, review states, evidence requirements, and correction loops. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for quality scoring, freshness, trust, review states, evidence requirements, and correction loops.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for quality scoring, freshness, trust, review states, evidence requirements, and correction loops.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeQualityDefined
- KnowledgeQualityIngested
- KnowledgeQualityIndexed
- KnowledgeQualityReviewed
- KnowledgeQualitySuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Quality has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.16 - Knowledge Security

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-016

Purpose

Define classification, tenant isolation, access policy, redaction, retention, and protected knowledge controls as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for classification, tenant isolation, access policy, redaction, retention, and protected knowledge controls. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for classification, tenant isolation, access policy, redaction, retention, and protected knowledge controls.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for classification, tenant isolation, access policy, redaction, retention, and protected knowledge controls.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeSecurityDefined
- KnowledgeSecurityIngested
- KnowledgeSecurityIndexed
- KnowledgeSecurityReviewed
- KnowledgeSecuritySuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Security has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.17 - Knowledge Events

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-017

Purpose

Define canonical events for ingestion, extraction, indexing, linking, approval, supersession, and retrieval as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for canonical events for ingestion, extraction, indexing, linking, approval, supersession, and retrieval. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for canonical events for ingestion, extraction, indexing, linking, approval, supersession, and retrieval.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for canonical events for ingestion, extraction, indexing, linking, approval, supersession, and retrieval.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeEventsDefined
- KnowledgeEventsIngested
- KnowledgeEventsIndexed
- KnowledgeEventsReviewed
- KnowledgeEventsSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Events has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.18 - Knowledge Testing

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-018

Purpose

Define testing strategy for ingestion, extraction, graph integrity, retrieval, quality, security, and versioning as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for testing strategy for ingestion, extraction, graph integrity, retrieval, quality, security, and versioning. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for testing strategy for ingestion, extraction, graph integrity, retrieval, quality, security, and versioning.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for testing strategy for ingestion, extraction, graph integrity, retrieval, quality, security, and versioning.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeTestingDefined
- KnowledgeTestingIngested
- KnowledgeTestingIndexed
- KnowledgeTestingReviewed
- KnowledgeTestingSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Testing has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.19 - Knowledge Acceptance

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-019

Purpose

Define acceptance gates for Knowledge System blueprint completeness, safety, retrieval readiness, and release evidence as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for acceptance gates for Knowledge System blueprint completeness, safety, retrieval readiness, and release evidence. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for acceptance gates for Knowledge System blueprint completeness, safety, retrieval readiness, and release evidence.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for acceptance gates for Knowledge System blueprint completeness, safety, retrieval readiness, and release evidence.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeAcceptanceDefined
- KnowledgeAcceptanceIngested
- KnowledgeAcceptanceIndexed
- KnowledgeAcceptanceReviewed
- KnowledgeAcceptanceSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Acceptance has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007
- MAL-V00-B010

- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.

Book 03.04.20 - Knowledge Roadmap

Version: v6 draft

Status: Draft

Document ID: MAL-V03-B03-04-020

Purpose

Define the implementation-facing roadmap for moving Knowledge System from blueprint to specifications and governed runtime behavior as part of the Mariam Knowledge System blueprint.

Scope

This document covers architecture-level responsibilities, contracts, data, events, controls, metrics, tests, and acceptance rules for the implementation-facing roadmap for moving Knowledge System from blueprint to specifications and governed runtime behavior. It does not implement system code or approve ungoverned data ingestion.

Responsibilities

- Establish the canonical boundary for the implementation-facing roadmap for moving Knowledge System from blueprint to specifications and governed runtime behavior.
- Preserve provenance, ownership, freshness, and review state for knowledge assets.
- Coordinate with Enterprise Core identity, storage, events, security, and observability services.
- Provide implementation-ready constraints for future knowledge specifications and AI Society retrieval behavior.

Inputs

- Enterprise Constitution knowledge, DNA, AI, security, quality, and governance principles.
- Master Roadmap Phase 04 Knowledge System requirements.
- Enterprise Core storage, event bus, identity access, security, and observability contracts.
- AI Society memory, validation, execution, and governance needs.

Outputs

- Blueprint contract for the implementation-facing roadmap for moving Knowledge System from blueprint to specifications and governed runtime behavior.
- Governed knowledge records, metadata, indexes, links, retrieval evidence, and quality signals.
- Traceable inputs for Volume 06 specifications and Volume 12 Knowledge System expansion.
- Acceptance evidence for ingestion, retrieval, security, versioning, and quality gates.

Interfaces

- Enterprise Core object manager, event bus, storage, security, and observability interfaces.
- AI Society agent memory, planning, validation, review, and governance interfaces.
- Future DNA Operating System memory and personalization interfaces.
- Future workflow, capability, connector, provider, and document ingestion interfaces.

Events

- KnowledgeRoadmapDefined
- KnowledgeRoadmapIngested
- KnowledgeRoadmapIndexed
- KnowledgeRoadmapReviewed
- KnowledgeRoadmapSuperseded

Storage

- Knowledge object records with source, owner, tenant, version, state, and classification metadata.
- Provenance records linking source material to extracted claims, entities, chunks, embeddings, and graph edges.
- Index and vector references that can be rebuilt from canonical source records.
- Review, correction, supersession, and retention history.

Security

- Apply tenant isolation and least privilege to knowledge objects, embeddings, indexes, and graph edges.
- Classify sensitive source material before extraction, indexing, embedding, or retrieval.
- Prevent generated or inferred knowledge from becoming authoritative without review.
- Audit protected reads, writes, exports, redactions, and retention changes.

Metrics

- Ingestion success rate, extraction quality, normalization accuracy, and indexing freshness.
- Retrieval relevance, citation coverage, answer grounding, and stale-result rate.
- Knowledge quality score, review backlog, correction rate, and supersession count.
- Protected access denial count, redaction rate, and retention policy compliance.

Tests

- Contract tests for knowledge object schemas, event envelopes, and storage interfaces.
- Retrieval evaluation tests for relevance, citation fidelity, filters, and tenant isolation.
- Security tests for classification, redaction, protected access, and embedding leakage risks.
- Versioning, rollback, re-indexing, freshness, and quality regression tests.

Acceptance Criteria

- Knowledge Roadmap has explicit source, provenance, security, and quality controls.
- Events, storage records, metrics, and tests are defined well enough for downstream specifications.
- Retrieval and AI usage remain grounded in approved or clearly labeled knowledge state.
- Traceability links this blueprint to Volume 00, Volume 02, Book 03.01, and Book 03.03.

Traceability

- MAL-V00-B007

- MAL-V00-B010
- MAL-V00-B011
- MAL-V01-B008
- MAL-V02-B006
- MAL-V03-B03-01-015
- MAL-V03-B03-03-010

Version History

Version	Date	Status	Notes
v6 draft	2026-06-29	Draft	Initial Knowledge System blueprint content for Mariam Architecture Library v6.